283. Why is $-2^4 \neq (-2)^4$?

284. Why is $-4^2 \neq (-4)^2$?

Self Check

ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
multiply integers.			
divide integers.			
simplify expressions with integers.			
evaluate variable expressions with integers.			
translate word phrases to algebraic expressions.			

ⓑ On a scale of 1–10, how would you rate your mastery of this section in light of your responses on the checklist? How can you improve this?

3.5 Solve Equations Using Integers; The Division Property of Equality

Learning Objectives

By the end of this section, you will be able to:

- Determine whether an integer is a solution of an equation
- Solve equations with integers using the Addition and Subtraction Properties of Equality
- Model the Division Property of Equality
- Solve equations using the Division Property of Equality
- Translate to an equation and solve

☑ **BE PREPARED** 3.10 Before you get started, take this readiness quiz.

Evaluate $x + 4$ when $x = -4$.
If you missed this problem, review Example 3.22.

☑ **BE PREPARED** 3.11 Solve: $y - 6 = 10$.
If you missed this problem, review Example 2.33.

☑ **BE PREPARED** 3.12 Translate into an algebraic expression *5 less than* x.
If you missed this problem, review Table 1.3.

Determine Whether a Number is a Solution of an Equation

In Solve Equations with the Subtraction and Addition Properties of Equality, we saw that a solution of an equation is a value of a variable that makes a true statement when substituted into that equation. In that section, we found solutions that were whole numbers. Now that we've worked with integers, we'll find integer solutions to equations.

The steps we take to determine whether a number is a solution to an equation are the same whether the solution is a whole number or an integer.



HOW TO

How to determine whether a number is a solution to an equation.

Step 1. Substitute the number for the variable in the equation.
Step 2. Simplify the expressions on both sides of the equation.
Step 3. Determine whether the resulting equation is true.
- If it is true, the number is a solution.
- If it is not true, the number is not a solution.

EXAMPLE 3.60

Determine whether each of the following is a solution of $2x - 5 = -13$:

ⓐ $x = 4$ ⓑ $x = -4$ ⓒ $x = -9$.

Solution

ⓐ Substitute 4 for x in the equation to determine if it is true.

	$2x - 5 = -13$
Substitute 4 for x.	$2(4) - 5 \stackrel{?}{=} -13$
Multiply.	$8 - 5 \stackrel{?}{=} -13$
Subtract.	$3 \neq -13$

Since $x = 4$ does not result in a true equation, 4 is not a solution to the equation.

ⓑ Substitute −4 for x in the equation to determine if it is true.

	$2x - 5 = -13$
Substitute -4 for x.	$2(-4) - 5 \stackrel{?}{=} -13$
Multiply.	$-8 - 5 \stackrel{?}{=} -13$
Subtract.	$-13 = -13$ ✓

Since $x = -4$ results in a true equation, -4 is a solution to the equation.

ⓒ Substitute −9 for x in the equation to determine if it is true.

	$2x - 5 = -13$
Substitute −9 for x.	$2(-9) - 5 \stackrel{?}{=} -13$
Multiply.	$-18 - 5 \stackrel{?}{=} -13$
Subtract.	$-23 \neq -13$

Since $x = -9$ does not result in a true equation, -9 is not a solution to the equation.

TRY IT 3.119 Determine whether each of the following is a solution of $2x - 8 = -14$:

ⓐ $x = -11$ ⓑ $x = 11$ ⓒ $x = -3$

TRY IT 3.120 Determine whether each of the following is a solution of $2y + 3 = -11$:

ⓐ $y = 4$ ⓑ $y = -4$ ⓒ $y = -7$

Solve Equations with Integers Using the Addition and Subtraction Properties of Equality

In Solve Equations with the Subtraction and Addition Properties of Equality, we solved equations similar to the two shown here using the Subtraction and Addition Properties of Equality. Now we can use them again with integers.

$$\begin{array}{ll} x+4=12 & y-5=9 \\ x+4-4=12-4 & y-5+5=9+5 \\ x=8 & y=14 \end{array}$$

When you add or subtract the same quantity from both sides of an equation, you still have equality.

Properties of Equalities

Subtraction Property of Equality	Addition Property of Equality
For any numbers a, b, c, if $a = b$ then $a - c = b - c$.	For any numbers a, b, c, if $a = b$ then $a + c = b + c$.

EXAMPLE 3.61

Solve: $y + 9 = 5$.

Solution

	$y+9=5$
Subtract 9 from each side to undo the addition.	$y+9-9=5-9$
Simplify.	$y=-4$

Check the result by substituting -4 into the original equation.

	$y+9=5$
Substitute −4 for y	$-4+9 \stackrel{?}{=} 5$
	$5=5$✓

Since $y = -4$ makes $y + 9 = 5$ a true statement, we found the solution to this equation.

TRY IT 3.121 Solve:

$y + 11 = 7$

TRY IT 3.122 Solve:

$y + 15 = -4$

EXAMPLE 3.62

Solve: $a - 6 = -8$

Solution

	$a - 6 = -8$
Add 6 to each side to undo the subtraction.	$a - 6 + 6 = -8 + 6$
Simplify.	$a = -2$
Check the result by substituting -2 into the original equation:	$a - 6 = -8$
Substitute -2 for a	$-2 - 6 \stackrel{?}{=} -8$
	$-8 = -8$ ✓

The solution to $a - 6 = -8$ is -2.

Since $a = -2$ makes $a - 6 = -8$ a true statement, we found the solution to this equation.

TRY IT 3.123 Solve:

$a - 2 = -8$

TRY IT 3.124 Solve:

$n - 4 = -8$

Model the Division Property of Equality

All of the equations we have solved so far have been of the form $x + a = b$ or $x - a = b$. We were able to isolate the variable by adding or subtracting the constant term. Now we'll see how to solve equations that involve division.

We will model an equation with envelopes and counters in Figure 3.21.

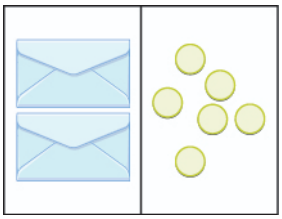


Figure 3.21

Here, there are two identical envelopes that contain the same number of counters. Remember, the left side of the workspace must equal the right side, but the counters on the left side are "hidden" in the envelopes. So how many counters are in each envelope?

To determine the number, separate the counters on the right side into 2 groups of the same size. So 6 counters divided into 2 groups means there must be 3 counters in each group (since $6 \div 2 = 3$).

What equation models the situation shown in Figure 3.22? There are two envelopes, and each contains x counters. Together, the two envelopes must contain a total of 6 counters. So the equation that models the situation is $2x = 6$.

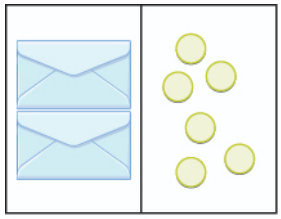


Figure 3.22

We can divide both sides of the equation by 2 as we did with the envelopes and counters.

$$\frac{2x}{2} = \frac{6}{2}$$
$$x = 3$$

We found that each envelope contains 3 counters. Does this check? We know $2 \cdot 3 = 6$, so it works. Three counters in each of two envelopes does equal six.

Figure 3.23 shows another example.

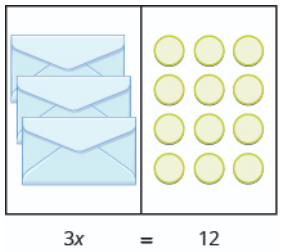


Figure 3.23

Now we have 3 identical envelopes and 12 counters. How many counters are in each envelope? We have to separate the 12 counters into 3 groups. Since $12 \div 3 = 4$, there must be 4 counters in each envelope. See Figure 3.24.

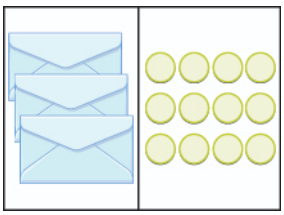


Figure 3.24

The equation that models the situation is $3x = 12$. We can divide both sides of the equation by 3.

$$\frac{3x}{3} = \frac{12}{3}$$
$$x = 4$$

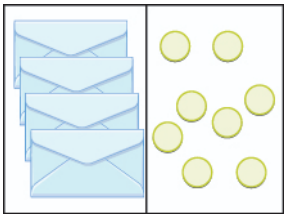
Does this check? It does because $3 \cdot 4 = 12$.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity "Division Property of Equality" will help you develop a better understanding of how to solve equations using the Division Property of Equality.

EXAMPLE 3.63

Write an equation modeled by the envelopes and counters, and then solve it.



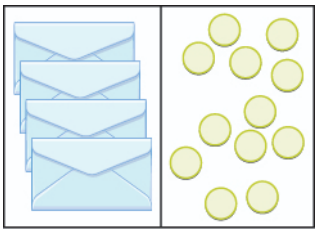
Solution

There are 4 envelopes, or 4 unknown values, on the left that match the 8 counters on the right. Let's call the unknown quantity in the envelopes x.

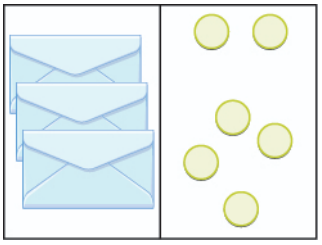
Write the equation.	$4x = 8$
Divide both sides by 4.	$\frac{4x}{4} = \frac{8}{4}$
Simplify.	$x = 2$

There are 2 counters in each envelope.

TRY IT 3.125 Write the equation modeled by the envelopes and counters. Then solve it.



TRY IT 3.126 Write the equation modeled by the envelopes and counters. Then solve it.



Solve Equations Using the Division Property of Equality

The previous examples lead to the Division Property of Equality. When you divide both sides of an equation by any nonzero number, you still have equality.

Division Property of Equality

For any numbers a, b, c, and $c \neq 0$,

If $a = b$ then $\frac{a}{c} = \frac{b}{c}$.

EXAMPLE 3.64

Solve: $7x = -49$.

Solution

To isolate x, we need to undo multiplication.

	$7x = -49$
Divide each side by 7.	$\frac{7x}{7} = \frac{-49}{7}$
Simplify.	$x = -7$

Check the solution.

	$7x = -49$
Substitute −7 for x.	$7(-7) \stackrel{?}{=} -49$
	$-49 = -49$✓

Therefore, -7 is the solution to the equation.

TRY IT 3.127 Solve:

$8a = 56$

TRY IT 3.128 Solve:

$11n = 121$

EXAMPLE 3.65

Solve: $-3y = 63$.

Solution

To isolate y, we need to undo the multiplication.

	$-3y = 63$
Divide each side by −3.	$\frac{-3y}{-3} = \frac{63}{-3}$
Simplify	$y = -21$

Check the solution.

	$-3y = 63$
Substitute –21 for y.	$-3(-21) \stackrel{?}{=} 63$
	$63 = 63$✓

Since this is a true statement, $y = -21$ is the solution to the equation.

TRY IT 3.129 Solve:

$-8p = 96$

TRY IT 3.130 Solve:

$-12m = 108$

Translate to an Equation and Solve

In the past several examples, we were given an equation containing a variable. In the next few examples, we'll have to first translate word sentences into equations with variables and then we will solve the equations.

EXAMPLE 3.66

Translate and solve: five more than x is equal to -3.

Solution

	five more than x is equal to -3
Translate	$x + 5 = -3$
Subtract 5 from both sides.	$x + 5 - 5 = -3 - 5$
Simplify.	$x = -8$

Check the answer by substituting it into the original equation.

$$x + 5 = -3$$
$$-8 + 5 \stackrel{?}{=} -3$$
$$-3 = -3$$✓

TRY IT 3.131 Translate and solve:

Seven more than x is equal to -2.

TRY IT 3.132 Translate and solve:

Eleven more than y is equal to 2.

EXAMPLE 3.67

Translate and solve: the difference of n and 6 is -10.

Solution

	the difference of n and 6 is -10
Translate.	$n-6=-10$
Add 6 to each side.	$n-6+6=-10+6$
Simplify.	$n=-4$

Check the answer by substituting it into the original equation.

$$n-6=-10$$
$$-4-6\stackrel{?}{=}-10$$
$$-10=-10\checkmark$$

TRY IT 3.133 Translate and solve:

The difference of p and 2 is -4.

TRY IT 3.134 Translate and solve:

The difference of q and 7 is -3.

EXAMPLE 3.68

Translate and solve: the number 108 is the product of -9 and y.

Solution

	the number of 108 is the product of -9 and y
Translate.	$108=-9y$
Divide by -9.	$\frac{108}{-9}=\frac{-9y}{-9}$
Simplify.	$-12=y$

Check the answer by substituting it into the original equation.

$$108=-9y$$
$$108\stackrel{?}{=}-9(-12)$$
$$108=108\checkmark$$

TRY IT 3.135 Translate and solve:

The number 132 is the product of -12 and y.

TRY IT 3.136 Translate and solve:

The number 117 is the product of -13 and z.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

One-Step Equations With Adding Or Subtracting (https://openstaxcollege.org/l/24onestepaddsub)

One-Step Equations With Multiplying Or Dividing (https://openstaxcollege.org/l/24onestepmuldiv)

SECTION 3.5 EXERCISES

Practice Makes Perfect

Determine Whether a Number is a Solution of an Equation

In the following exercises, determine whether each number is a solution of the given equation.

285. $4x - 2 = 6$
ⓐ $x = -2$ ⓑ $x = -1$
ⓒ $x = 2$

286. $4y - 10 = -14$
ⓐ $y = -6$ ⓑ $y = -1$
ⓒ $y = 1$

287. $9a + 27 = -63$
ⓐ $a = 6$ ⓑ $a = -6$
ⓒ $a = -10$

288. $7c + 42 = -56$
ⓐ $c = 2$ ⓑ $c = -2$
ⓒ $c = -14$

Solve Equations Using the Addition and Subtraction Properties of Equality

In the following exercises, solve for the unknown.

289. $n + 12 = 5$

290. $m + 16 = 2$

291. $p + 9 = -8$

292. $q + 5 = -6$

293. $u - 3 = -7$

294. $v - 7 = -8$

295. $h - 10 = -4$

296. $k - 9 = -5$

297. $x + (-2) = -18$

298. $y + (-3) = -10$

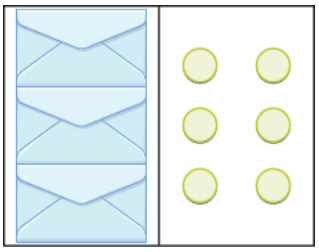
299. $r - (-5) = -9$

300. $s - (-2) = -11$

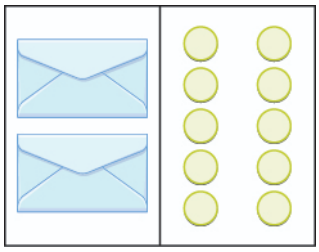
Model the Division Property of Equality

In the following exercises, write the equation modeled by the envelopes and counters and then solve it.

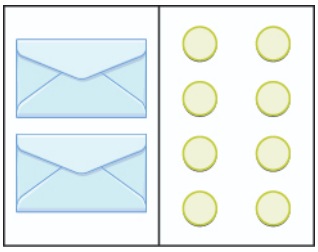
301.



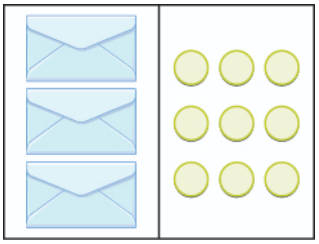
302.



303.



304.



Solve Equations Using the Division Property of Equality

In the following exercises, solve each equation using the division property of equality and check the solution.

305. $5x = 45$

306. $4p = 64$

307. $-7c = 56$

308. $-9x = 54$

309. $-14p = -42$

310. $-8m = -40$

311. $-120 = 10q$

312. $-75 = 15y$

313. $24x = 480$

314. $18n = 540$

315. $-3z = 0$

316. $4u = 0$

Translate to an Equation and Solve

In the following exercises, translate and solve.

317. Four more than n is equal to 1.

318. Nine more than m is equal to 5.

319. The sum of eight and p is -3.

320. The sum of two and q is -7.

321. The difference of a and three is -14.

322. The difference of b and 5 is -2.

323. The number −42 is the product of −7 and x.

324. The number −54 is the product of −9 and y.

325. The product of -15 and f is 75.

326. The product of −18 and g is 36.

327. −6 plus c is equal to 4.

328. −2 plus d is equal to 1.

329. Nine less than m is −4.

330. Thirteen less than n is -10.

Mixed Practice

In the following exercises, solve.

331. ⓐ $x + 2 = 10$
ⓑ $2x = 10$

332. ⓐ $y + 6 = 12$
ⓑ $6y = 12$

333. ⓐ $-3p = 27$
ⓑ $p - 3 = 27$

334. ⓐ $-2q = 34$
ⓑ $q - 2 = 34$

335. $a - 4 = 16$

336. $b - 1 = 11$

337. $-8m = -56$

338. $-6n = -48$

339. $-39 = u + 13$

340. $-100 = v + 25$

341. $11r = -99$

342. $15s = -300$

343. $100 = 20d$

344. $250 = 25n$

345. $-49 = x - 7$

346. $64 = y - 4$

Everyday Math

347. **Cookie packaging** A package of 51 cookies has 3 equal rows of cookies. Find the number of cookies in each row, c, by solving the equation $3c = 51$.

348. **Kindergarten class** Connie's kindergarten class has 24 children. She wants them to get into 4 equal groups. Find the number of children in each group, g, by solving the equation $4g = 24$.

Writing Exercises

349. Is modeling the Division Property of Equality with envelopes and counters helpful to understanding how to solve the equation $3x = 15$? Explain why or why not.

350. Suppose you are using envelopes and counters to model solving the equations $x + 4 = 12$ and $4x = 12$. Explain how you would solve each equation.

351. Frida started to solve the equation $-3x = 36$ by adding 3 to both sides. Explain why Frida's method will not solve the equation.

352. Raoul started to solve the equation $4y = 40$ by subtracting 4 from both sides. Explain why Raoul's method will not solve the equation.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
determine whether an integer is a solution of an equation.			
solve equations with integers using the addition and subtraction properties of equality.			
model division property of equality.			
solve equations using the division property of equality.			
translate to an equation and solve.			

ⓑ *Overall, after looking at the checklist, do you think you are well-prepared for the next Chapter? Why or why not?*